

ISAAC FATOKUN

Software Engineer — .NET, Systems Integration & Reliability

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SUMMARY

Software Engineer with 4+ years of experience across asynchronous message processing, systems integration, and transactional workflows. Experienced across modern .NET and legacy .NET Framework 4.8, with Entity Framework Core, SQL Server, and queue-based integration in multithreaded services. Strong background in message-driven processing, background and scheduled work, resilient integration with external dependencies, observability, and automated testing, with a particular focus on reliability and performance in production.

EXPERIENCE

Software Engineer — Core Laboratory Microbiology Systems <i>Eurofins Scientific</i>	Nantes, France Sep 2023 – Present
- Contribute across 14+ production services in a service-oriented platform, spanning user-facing applications and multithreaded services processing queued instrument messages through an enterprise service bus, across modern .NET and .NET Framework 4.8. Own the background result-import pipeline: process bulk messages (~100 results each), resolve results against existing data, link them to the correct sample, test, and attempt, and persist outcomes with failure reasons. Supports fallback matching for partial identifiers, reducing up to 6 hours of manual processing for labs. Implemented parameter coherence validation in the next-generation calculation engine, enforcing consistency across derived and reported values with defined precedence and dependency ordering. Introduced Elasticsearch after relational queries reached performance limits at volume; owned index design, reindexing, query tuning, and rollout. Engineered resilient external-provider access with distributed caching, scheduled preloading, semaphore-guarded refresh, and Polly retry, backoff, and circuit breaking. Implemented service-to-service authentication using OAuth 2.0 client credentials, Microsoft Entra ID, client certificates, and Windows Integrated Authentication. Monitored production services through Grafana dashboards and structured NLog logging, using application telemetry to diagnose issues and support faster incident resolution; resolved 20+ defects in the core analysis application. Resolved concurrency defects, including synchronization-context deadlocks caused by blocking on async work; restructured execution paths and applied appropriate async patterns. Applied TDD with NUnit, MSTest, and Moq across business-critical workflows; ran FitNesse acceptance testing and facilitated sessions on concurrency and async/await within a 30+ member Agile team.	
Associate Software Engineer — Platform Research & Development	Mar 2023 – Aug 2023
- Delivered a containerised maintenance application in six months (Docker), from requirements through to production, replacing a workflow in which reference data was written and edited by hand. - Moved that data behind a validated, access-controlled application with scoped hosting, removing the manual direct-edit path and the data-integrity and security exposure it carried. Tech Stack: C#, Angular, HTML, CSS, TypeScript, .NET Core, .NET Framework 4.8, ASP.NET Core, Entity Framework Core, SQL Server (T-SQL), message queues, enterprise service bus, Hangfire, Polly, Elasticsearch, NLog, Grafana, Docker, Azure DevOps, NUnit, MSTest, Moq, Git	
Software Engineering Technician - Semantic Systems <i>École des Mines de Saint-Étienne</i>	Saint-Etienne, France Oct 2022 – Jan 2023
First author of a modular digital twin architecture for smart buildings, integrating five heterogeneous sources — BIM models, an employee directory, a KNX building-automation network, live MQTT sensor data, and historical logs — into a single navigable model. 2nd place, Best Paper, LDAC 2023. - Reverse-engineered proprietary building-automation configuration files to derive device topology, properties, and communication objects, modelled against W3C standards. - Developed a pre-commit linting system for semantic web applications, identifying 85% more syntax errors pre-deployment. Tech Stack: Python, RDF/OWL, SPARQL, KNX, MQTT, W3C semantic web standards, Git	

Palaiseau, France

Software Engineering Intern

EDF R&D / Télécom SudParis

April 2022 – Aug 2022

- **Built and shipped a modular data model** for smart buildings, improving tenant data management across 75% of targeted inefficient buildings while meeting EU energy compliance requirements.

Tech Stack: Python, RDF/OWL, Git

EDUCATION

École des Mines de Saint-Étienne / Université Jean Monnet

France

Master of Science – Computer Science - *Cyber-Physical-Social Systems*

Sep 2021 – Aug 2023

- Coursework: Distributed Systems, Software Architecture, Cloud Engineering, Data Modelling

Wesley University

Ondo, Nigeria

Bachelor of Science – Computer Science

Mar 2015 – Feb 2020

PUBLICATIONS

Modular Knowledge Integration for Smart Building Digital Twins — Fatokun, I. et al. LDAC'23, Linked Data in Architecture and Construction Workshop, Matera, Italy, June 2023. 2nd place, Best Paper. CEUR Proceedings · HAL: emse-04086955

TECHNICAL SKILLS

Languages & Backend: C#, .NET 6/8, .NET Framework 4.8, ASP.NET Core, Python, TypeScript, Go

Data & Search: SQL Server, PostgreSQL, Entity Framework Core, Elasticsearch, transactions, indexing, query optimisation

Messaging & Integration: REST APIs, Enterprise Service Bus, message queues, asynchronous and bulk processing, Hangfire, service-to-service integration

Reliability & Concurrency: Multithreading, thread safety, optimistic concurrency, caching, retries, circuit breakers, graceful shutdown

Security: OAuth 2.0, Microsoft Entra ID, client certificates, Windows Authentication

Observability & Delivery: Grafana, NLog, Docker, Azure, Azure DevOps, CI/CD, Git

Testing: TDD, Unit & Integration Testing, FitNesse, NUnit, MSTest, Moq

CERTIFICATIONS

LFS250: Kubernetes and Cloud Native Essentials — The Linux Foundation (May 2026) — [Certificate](#) **DELFI B1** — France Éducation International (Dec 2025)

LANGUAGE PROFICIENCY & INTERESTS

Languages: English (Native) • French (B1 - DELFI)

Interests: Teaching, Music, AI Research, Cooking